Label Propagation and Contrastive Regularization for Semisupervised Semantic Segmentation of Remote Sensing Images

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***Abstract*— Remarkable progress based on deep neural networks has been achieved in the semantic segmentation of remote sensing images (RSIs). However, pixel-level labeling is expensive for RSIs. Semisupervised semantic segmentation becomes an alternative approach to reduce the cost of annotation, and it is crucial to utilize efficiently a large number of unlabeled data. Nevertheless, inevitably, there is an unbalanced class distribution between labeled and unlabeled data in a remote sensing scene. Existing semisupervised methods train unlabeled images in isolation from labeled images and only learn reliable pixel pseudo-labels, leading to underutilization of unlabeled images. This article proposes a novel semisupervised semantic segmentation approach based on label propagation and contrastive regularization for RSIs. Specifically, the unlabeled images are augmented by randomly copy-pasting the class regions from labeled images. A prototype feature constraint module is used to enforce the constraint on the pixel features of unlabeled images relying on the prototype features from labeled images, achieving feature alignment on the entire dataset. Furthermore, we present the region contrastive learning (RCL) module that guides the model to learn feature consistency under different perturbations and compact feature representations over class regions on unlabeled images. Extensive experimental results on multiple remote sensing datasets demonstrate that our proposed approach achieves superior performance compared with state-of-the-art semisupervised semantic segmentation methods.**

***Index Terms*— Contrastive regularization, label propagation, remote sensing images (RSIs), semantic segmentation, semisupervised learning (SSL).**

I. Introduction

SEMANTIC segmentation as the fundamental task for remote sensing images (RSIs) interpretation plays a relevant role in widespread applications, such as urban planning, land-cover surveillance, and traffic management [1], [2], [3], [4]. Deep convolutional neural networks [5] have facilitated significant advances in semantic segmentation of RSIs [6], [7], [8], [9], [10], [11], [12]. These supervised learning approaches highly demand massive fine-grained labeled images to obtain generalized models, and however, the pixelwise labeling is costly. Especially in remote sensing scenes with the impact of factors such as ambiguous spectral imaging and poor lighting conditions, sufficient manual annotation is time-consuming and laborious. Semisupervised semantic segmentation has emerged as an alternative learning strategy to relieve this problem, which trains the segmentation network leveraging a few labeled data and a large amount of unlabeled data.

Current pipelines for semisupervised learning (SSL) on semantic segmentation typically focus on consistency regularization (CR) and self-training (ST) based on pseudo-labels. The core concept of CR lies in generating invariant predictions for the input of unlabeled images under different perturbations (e.g., augmented images, feature perturbations, and network perturbations) [13], [14], [15], [16]. CR methods separate samples by spotting low-density regions based on the cluster assumption [13], [17]. Ouali et al. [15] proposed cross-consistency training that imposes different perturbations on the output features of the encoder and makes outputs of multiple decoders uniform. Nevertheless, the selection of perturbations is cumbersome and those methods suffer from weak universality. ST usually utilizes the model trained with labeled samples to provide predictions on unlabeled samples and then retrains the unlabeled samples with pseudo-labels as ground truth [18], [19], [20], [21]. How to train pseudo-labels is a frequent concern. Feng et al. [19] proposed dynamic mutual training that uses disagreements between models to build the robust loss function with respect to pseudo-labels by predicting the confidence. However, ST-based methods are hard to guarantee the generation of reliable pseudo-labels, and it commonly causes the model to be biased toward easy categories [21].

Recently, FixMatch proposed by Sohn et al. [22] unifies the dominant idea of ST and CR, which has a remarkable performance in the classification task. It uses pseudo-labels generated from weakly augmented samples to supervise strongly augmented samples, which allows the model to learn invariance under different perturbations and expand the training samples with pseudo-labels. Consequently, many FixMatch-like semisupervised semantic segmentation methods

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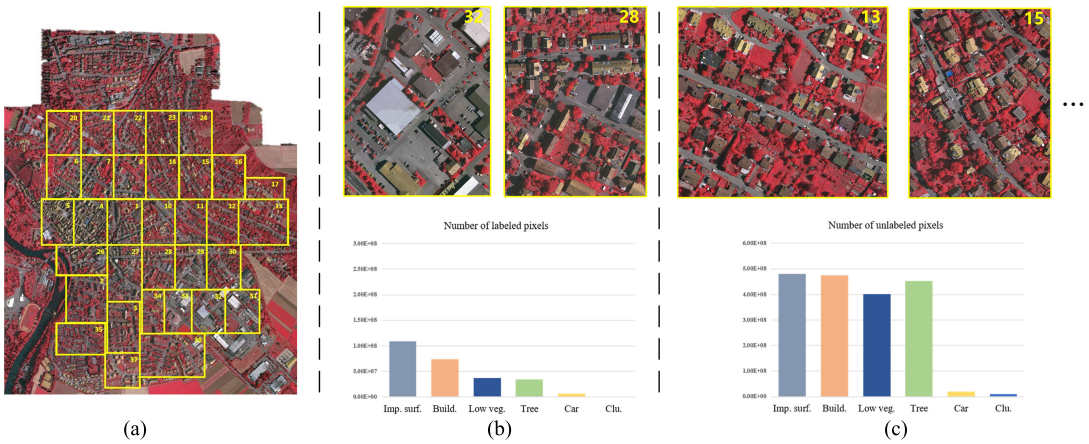


Fig. 1. ISPRS Vaihingen dataset used for training. With only 1/8 labeled images, the labeled images are the 28th and 32nd scenes randomly selected among the 16 training images, with the remaining 14 as the unlabeled dataset. There is a clear imbalance in the class distribution between labeled and unlabeled images, especially for low vegetation and tree. (a) ISPRS Vaihingen dataset. (b) Labeled dataset. (c) Unlabeled dataset.

have been studied, and they focus on effective pseudo-labels and well-designed data augmentation [23], [24], [25], [26]. However, how to leverage effectively the unlabeled data is the key to semisupervised semantic segmentation, and these methods have the following limitations when used directly on remote sensing data.

The process of training unlabeled data is isolated from labeled data, which yields suboptimal performance in remote sensing scenes. This is attributed to the fact that RSIs cover large-scale scenes with wide variations in the distribution of ground elements. It is hard to avoid that the labeled dataset and unlabeled dataset would own unbalanced class distribution. Fig. 1 shows the distribution of samples with only 1/8 labeled images in the ISPRS Vaihingen training set. Since the difference in scenes, we can see that pixels of low vegetation and tree account for a smaller proportion in the annotated industrial area images compared to the unannotated residential area images. The model pretrained on inadequate labeled data has poor generalization and is prone to produce dispersed features for unlabeled data. Thus, segregated training for annotated and unannotated data makes low utilization of unlabeled images, especially for classes with few labeled samples. We argue that joint training of labeled and unlabeled images is essential to address the imbalance in class distribution.

The unlabeled data are trained with reliable pixel pseudo-labels only that yield prediction confidences higher than the predefined threshold. In fact, it is difficult to achieve category separability over large amounts of unlabeled data using only pixel-level pseudo-labels as supervision in complex remote sensing scenarios. In Fig. 2(a), the model trained with reliable pseudo-labels generates low prediction confidence values for the unlabeled image in the object boundary and regions with inferior generalization (e.g., tree). Meanwhile, Fig. 2(b) presents its feature space encoded by the model, where a clear category confusion exists. Inspired by the application of contrastive learning to SSL [27], [28], [29], we propose to develop region–structure-based feature representation learning on unlabeled images. In addition to pixel-level consistency under different perturbations [27], [28], we also promote the

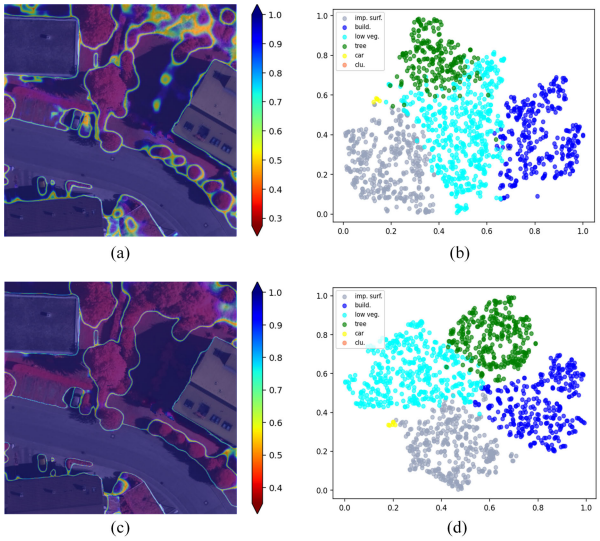


Fig. 2. Pseudo-label confidence map and t-SNE [30] visualization of features. (a) and (b) Model trained with reliable pseudo-labels. (c) and (d) Model with our method.

same class regions into similar feature representations and the differences across class regions. In Fig. 2(c) and (d), our method produces pseudo-label predictions with high confidence and also has more significant feature separability and category uniformity.

To tackle the above problems, we propose the label propagation and contrastive regularization framework for semisupervised semantic segmentation of RSIs. The framework employs a teacher–student scheme following mean teacher (MT) [31] with three innovative components, as shown in Fig. 3. First, random copy-paste (RCP) is introduced, which performs from labeled images to unlabeled images so that certain insufficiently labeled classes with their ground truth are blended into unlabeled images for training. Compared to the existing data augmentations for SSL [25], [32], RCP accomplishes the explicit label propagation and forms new samples with better semantic boundaries. Then, we propose a prototype feature constraint (PFC) module. The prototypes, i.e., the

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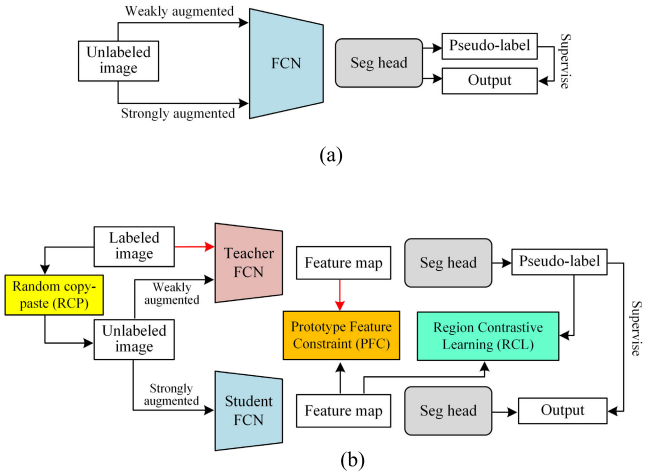


Fig. 3. Pipelines of semisupervised semantic segmentation. (a) Current FixMatch-like methods are intended to use pseudo-labels generated from weakly augmented samples to supervise strongly augmented samples. (b) Our architecture builds on the MT schema with novel RCP, PFC, and RCL modules.

classwise feature centroids derived from the teacher model using labeled data, are stored in a prototype memory bank (PMB). Given an unlabeled image, we compute the distance between its feature and elements in the PMB and train the student model with similar class features. The PFC module utilizes exact representations from labeled data to constrain unlabeled data so that features of the unlabeled data are aligned to the labeled data via the implicit label propagation. Finally, we design a region contrastive learning (RCL) module to perform contrastive regularization based on class regions guided by a pseudo-label graph constructed with pseudo-labels' confidence. The RCL module combines the ideas of CR and RCL to train invariant features under perturbations and compact class features for unlabeled data.

In summary, our contributions are summarized as follows.

1) We present a unified framework based on label propagation and contrastive regularization for semisupervised semantic segmentation of RSIs. Our framework bridges the unbalanced class distribution of labeled and unlabeled data to take more full advantage of unlabeled data under a few annotated data.
2) The PFC module is adopted for feature alignment throughout the dataset, which imposes a constraint on the pixel features of unlabeled images with the exact prototype representations from labeled images. The RCP provides boundary-accurate instances for unlabeled images.
3) The RCL module performs contrastive regularization on feature representations of unlabeled data through structural knowledge of pseudo-label graphs. It lets the model learn invariant features under perturbations and compact class features.

The rest of this article is arranged as follows. Section II introduces the related work of our research. Section III describes the proposed semisupervised semantic segmentation framework. Experiments and discussions are given in Section IV. Finally, Section V summarizes our work.

II. Related Works

We present related works, including semantic segmentation, semisupervised semantic segmentation, and SSL in RSIs.

A. Semantic Segmentation

The purpose of semantic segmentation is to assign a matching class label to each pixel in an image. Some of the current segmentation algorithms use superpixels as processing units [33], [34], [35], and others are pixel-level [6], [9], [36]. The superpixel-based approaches generally consist of the initial oversegmentation and the optimal merging steps. Ma et al. [34] proposed an end-to-end trainable network to synchronously update the shapes of superpixels and merging strategies for synthetic aperture radar (SAR) image segmentation. Wang et al. [35] proposed a segmentation method using minimum spanning tree incorporating superpixels used in high-resolution RSI. The performance of such methods is usually limited by the quality of superpixel extraction.

In recent years, pixel-level segmentation models designed based on fully convolutional network (FCN) [37] have made impressive progress in both the natural [38], [39], [40], [41] and remote sensing scene [6], [9], [11], [12], [36], [42]. Most of the subsequent approaches are studied with contextual information and edge refinement. Concretely, the works exploiting context information focus on two main aspects: multiscale spatial context and relational context. For instance, PSPNet [38] and ASPP [39] are concerned with the former. As a classic network exploring the relational context, OCRNet [40] computes the relationship between pixels and object regions, and then, the aggregated object context representation enhances the representation of each pixel. To model the context dependence of RSIs with complex spectra, Niu et al. [6] proposed HMANet to capture global correlations adaptively from spatial, channel, and category perspectives. There are also some efforts that are concerned with improving the quality of the segmentation boundaries. For instance, Gated-SCNN [41] designs an independent shape branch in which the gated convolutional layer extracts the edge information. BAS4Net proposed by Sun et al. [9] aims at improving the refinement of boundaries by a boundary attention module for high-resolution RSIs. In general, these excellent segmentation models are mainly based on fully supervised training and depend on costly and refined data annotation.

B. Semisupervised Semantic Segmentation

SSL provides a promising alternative to alleviate the requirement of expensive annotations for model training. In particular, many works have designed semisupervised training strategies for the natural scene to decrease the overhead of pixel labeling for semantic segmentation.

One of the ways is CR, which forces the predictions or features with various perturbations to remain consistent. Such methods are based on the cluster assumption that the decision function lies in the low-density region via prediction invariance with perturbations, such as temporal ensembling [43], MT [31], and cutmix-seg [13]. In cross-consistency training, Ouali [15] considered that for semantic segmentation, the

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low-density region for decision is present in the feature output, thus designing the consistent training of multiple decoders with feature perturbations. Lai et al. [16] suggested that contextual augmentation is more effective for training robust representations of the segmentation model compared to low-level perturbations. They propose to make instances learn consistent features under different contexts. However, it is cumbersome to select and add the appropriate perturbations. Well-designed augmentations might be sensitive to the training data.

Another way is ST or self-learning, which utilizes the model trained with labeled samples to provide predictions on unlabeled samples, then retrains the unlabeled samples with pseudo-labels as ground truth during multiple stages. However, how to select pseudo-labels for training is a frequent concern. Hung et al. [44] designed an FCN discriminator for distinguishing between ground truth and prediction probability maps and calculated confidence maps for segmentation prediction of unlabeled images. Feng et al. [19] proposed to build the robust dynamic loss to the pseudo-label using inconsistent models' predictions. Yang et al. [20] suggested injecting strong data augmentations for the ST paradigm to mitigate the overfitting of the model to the noisy labels. Nevertheless, ST-based methods are hard to guarantee the generation of accurate pseudo-labels and the misinformation in the pseudo-labels is prone to accumulate during the iterative training.

PseudoSeg [23] follows the FixMatch [22] scheme using pseudo-labels generated from weakly augmented samples to supervise strongly augmented samples. Subsequent FixMatch-like methods focus on effective pseudo-labels and well-designed data augmentation [23], [24], [25], [26] for semantic segmentation. Those approaches not only obtain consistent predictions under different perturbations but also expands the training samples with pseudo-labels to learn rich semantic features. Chen et al. [24] proposed to perform cross-supervision between the predictions of two segmentation networks whose initialization weights are different. ClassMix [25] mixes unlabeled images for data augmentation and trains the model by the combination of MT [31] and pseudo-labeling. Hu et al. [26] presented the adaptive data augmentation and data sampling strategy considering the long-tail label distribution. Wang et al. [45] proposed to select unreliable pixels via the entropy of predictions and train the model with their negative sample property. However, the training of pseudo-labels is separated from the labeled data in these approaches, resulting in scattered features across the dataset. Hence, we explore from a limited number of truth annotations to maximize the propagation of labeled knowledge and the utilization of unlabeled data.

C. SSL in RSIs

RSIs are influenced by various factors such as spectral imaging and light conditions, so detailed data labeling requires extensive expert knowledge and even on-the-spot surveying. In order to decrease the amount of human intervention required to train models, weakly supervised learning methods have been proposed for semantic segmentation [46], [47], [48]. In fact, massive amounts of remote sensing data are being acquired every day. It is highly valuable to utilize massive unlabeled images to assist in the training of models.

Recently, some SSL methods have emerged for RSIs. In [9], the proposed BAS4Net uses the fully convolutional discriminative network to derive the confidence of pseudo-labels, enabling the model to learn semantic information from unlabeled images. Wang et al. [49] suggested a method by combining CR under perturbations and average update of pseudo-label features. In the work [32], they propose RanPaste to use random boxes to fuse labeled and unlabeled data and utilize pseudo-labels controlled by thresholds. In contrast, we are inspired by ClassMix [25] to augment unlabeled images with labeled object instances that preserves rich boundary information. Li et al. [50] proposed consistency ST, which uses generative adversarial network (GAN) to generate pixel-level confidence for weighted unsupervised loss and imposes consistency constraints on the generators with random perturbations.

Although these methods succeed in semisupervised semantic segmentation on RSIs, they fail to sufficiently consider the data distribution characteristics and refinement of the features. We improve the limitations of the existing methods on RSIs and propose a semisupervised strategy that takes full use of unlabeled data.

III. Methodology

We present the details of the proposed method, including overall framework, RCP, PFC, and RCL modules.

A. Problem Setting

We anticipate effectively leveraging both labeled and unlabeled data to train a semantic segmentation network. Foremost, the semisupervised semantic segmentation problem is defined as follows. The dataset available for our training is composed of two portions: the labeled dataset $\mathcal{D}_l = \{(\mathbf{x}_i^l, \mathbf{y}_i^l)\}_{i=1}^{N_l}$, where $\mathbf{y}_i^l$ is corresponding annotation for image $\mathbf{x}_i^l$ and the unlabeled dataset $\mathcal{D}_u = \{\mathbf{x}_i^u\}_{i=1}^{N_u}$.

Fig. 4 shows an overview of the proposed framework. Overall, we employ the MT [31] training strategy and there are a teacher model f_ξ and a student model f_θ, which have the same architecture. The teacher model creates robust pseudo-labels from unlabeled images. At each training iteration, student network f_θ is optimized by gradient descent following the designed optimization objective, while weights ξ of teacher network f_ξ are updated according to the exponential moving average (EMA) scheme by weights θ of student model f_θ as follows:

$$\xi_t = \tau \xi_{t-1} + (1-\tau)\theta_t \tag{1}$$

where τ is the decay parameter $\tau \in [0, 1]$. In this way, a teacher model serves as a temporal ensemble for student model along the training iteration t, leading to robust predictions on unlabeled images.

The supervised loss on every labeled image is calculated by the cross-entropy loss, which is denoted as

$$L_{\text{sup}} = \frac{1}{|B_l|} \sum_{(\mathbf{x}_i^l, \mathbf{y}_i^l) \in B_l} \ell_{ce}\left(f_\theta\left(\mathbf{x}_i^l; \theta\right), \mathbf{y}_i^l\right) \tag{2}$$

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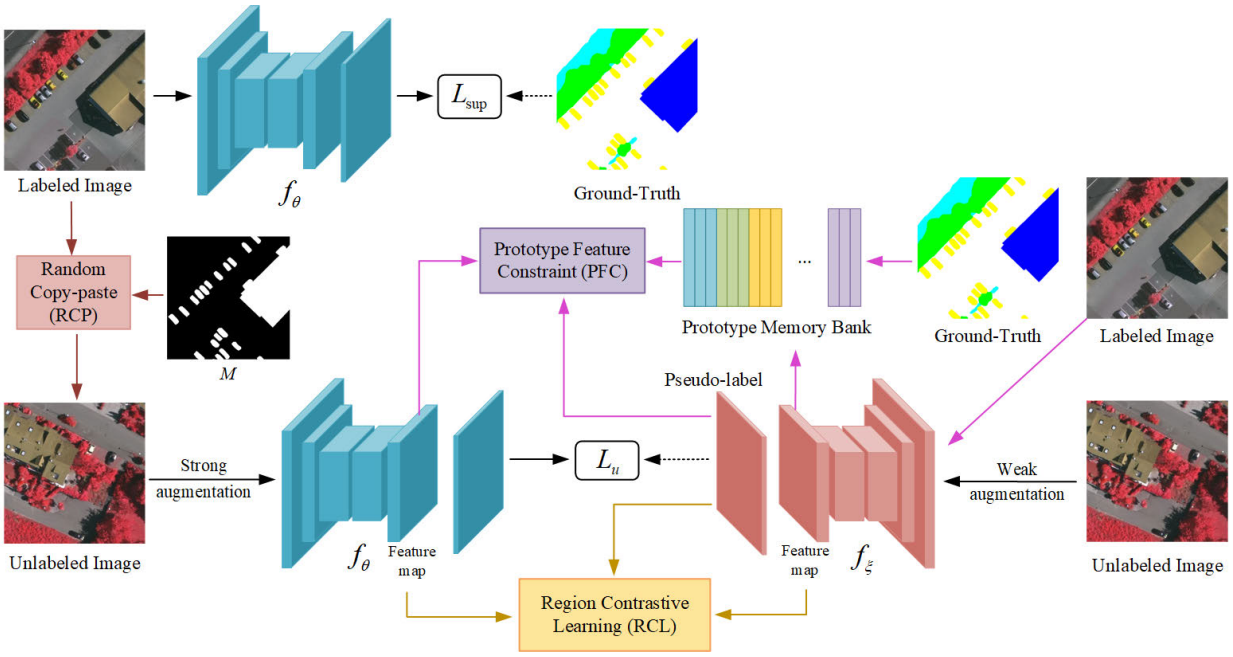


Fig. 4. Overview of the proposed semisupervised semantic segmentation framework with label propagation and contrastive regularization. The key components include three parts: an RCP for augmenting unlabeled images, a PFC module for constraints on unlabeled pixel features using labeled prototypes, and a region contrast learning module for training compact class features of unlabeled images.

where there are B_l labeled images in a mini-batch. For each unlabeled image, it is taken into the teacher model to obtain the pixel-level pseudo-label at each training iteration. Then, we compute the pseudo-labels loss by a cross-entropy loss function:

$$L_u = \frac{1}{|B_u|} \sum_{\mathbf{x}_i^u \in B_u} \ell_{ce}\left(f_\theta\left(\mathbf{x}_i^u; \theta\right), \hat{\mathbf{y}}_i^u\right) \tag{3}$$

where $\hat{\mathbf{y}}_i^u$ is the pseudo-label of unlabeled image $\mathbf{x}_i^u$ following the equation $\hat{\mathbf{y}}_i^u = \arg\max f_\xi(\mathbf{x}_i^u)$. Also, there are B_u unlabeled images in a mini-batch.

In Fig. 4, labeled images are applied to optimize the training of unlabeled images in addition to updating the model by a cross-entropy loss. Specifically, the mask M is generated for fusing the labeled category regions with the unlabeled images by RCP, which provides boundary-accurate instances for unlabeled images. The PFC module stores a prototype feature bank generated by a teacher model with the labeled dataset. The unlabeled images are then constrained by prototype features belonging to the same class from the labeled dataset to achieve that features of the unlabeled data are aligned with the labeled data. Furthermore, in order to learn compact class features of unlabeled data, the designed RCL module trains the region–structure relationships of the feature map guided by pseudo-labels via injecting different perturbations (data augmentation).

B. DA Through RCP

Several previous studies have confirmed the facilitative effect of suitable data augmentation on SSL [13], [16], [25]. Among them, ClassMix [25] mixes unlabeled images using pseudo-labels for respecting semantic boundaries. Our proposed RCP can be regarded as an extension of ClassMix

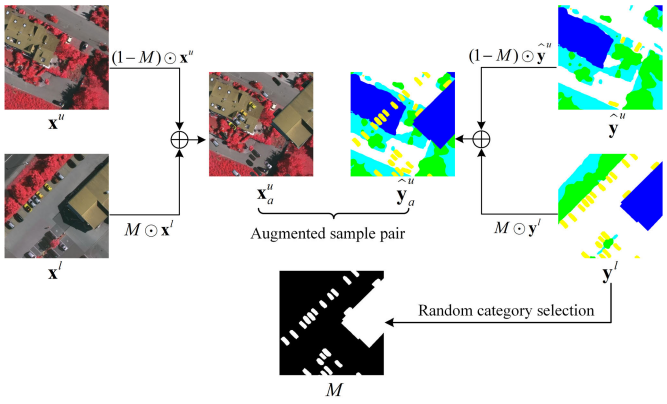


Fig. 5. Process of RCP. The mask M generated from ground truth fuses labeled class regions into unlabeled images to synthesize new trainable sample pairs.

for RSIs. Due to the inevitable problem of unbalanced data distribution in remote sensing scenes, we aim to improve the training effect on unlabeled images by explicitly making full use of labeled pixel annotations and propagating label information to unlabeled images. Fig. 5 shows the process of RCP. We randomly sample the labeled image $\mathbf{x}^l$ with its corresponding ground truth $\mathbf{y}^l$ and the unlabeled image $\mathbf{x}^u$ with pseudo-label $\hat{\mathbf{y}}^u$ generated by the teacher model. The binary mask M is generated by randomly selecting part of the category regions in the ground truth $\mathbf{y}^l$. Assume that there are C categories in the image $\mathbf{x}^l$ and we randomly select a subset c with $C/2$ categories. The binary mask M is generated as follows:

$$M_{(i,j)} = \begin{cases} 1, & \text{if } \mathbf{x}^l_{(i,j)} \in c \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

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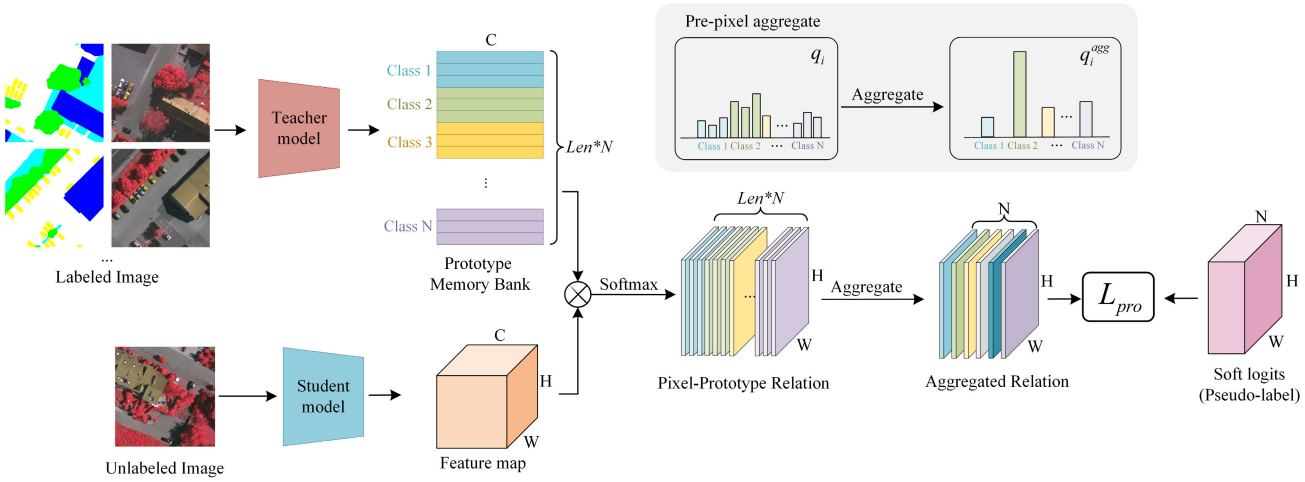


Fig. 6. Illustration of the PFC module. We store the labeled prototype features in the PMB and compute the relation between pixel features of unlabeled images and prototype features, and then, the aggregated relation is constrained by the soft pseudo-label.

Then, this mask is used to copy-paste randomly selected partial categories from the labeled image to the unlabeled image. The augmented unlabeled image is denoted as

$$\mathbf{x}_a^u \leftarrow M \odot \mathbf{x}^l + (1 - M) \odot \mathbf{x}^u \tag{5}$$

where $\odot$ indicates the elementwise product. Meanwhile, the corresponding pseudo-label is produced by

$$\hat{\mathbf{y}}_a^u \leftarrow M \odot \mathbf{y}^l + (1 - M) \odot \hat{\mathbf{y}}^u. \tag{6}$$

Hence, $\mathbf{x}_a^u$ and $\hat{\mathbf{y}}_a^u$ are taken as an augmented sample pair for training. It can be seen that the category regions from the labeled image $\mathbf{x}^l$ have more accurate boundary class annotations. Also, the generated coarse pseudo-labels are replaced with accurate annotations in covered areas. In each iteration of training, the RCP is performed with a probability of 0.5. The detailed analysis of the effectiveness is presented in Section IV.

C. PFC Module

Previous semisupervised methods use only limited labeled data to obtain a preliminary model, generating pseudo-labels to train the unlabeled data. The separated training way has worse performance under the large-scale scene of RSIs. The geographical characteristics of RSIs vary widely with regions, and it is hard to avoid the phenomenon of class unbalance. This results in a failure to learn consistent class representations on both labeled and unlabeled data. To alleviate the problem and fully use the high-quality labeled data, we propose PFC with the process shown in Fig. 6.

First, the teacher model stores a memory bank of prototype features from labeled data, named PMB. For the input labeled image $\mathbf{x}^l$, we define the prototype of class k as the mean of features of all pixels that fall into class k in image $\mathbf{x}^l$. Formally, given the feature map $f_{\xi^-}(\mathbf{x}^l) \in \mathbb{R}^{C\times H\times W}$ in which f_{ξ^-} is the teacher's encoder, the calculation of the prototype $\eta^{(k)}$ of class k can be expressed as

$$\eta^{(k)} = \frac{\sum_i f_{\xi^-}(\mathbf{x}^l) * \mathbb{1}[\hat{\mathbf{y}}_i^l == k]}{\sum_i \mathbb{1}[\hat{\mathbf{y}}_i^l == k]} \tag{7}$$

where $\hat{\mathbf{y}}^l = \arg\max f_{\xi}(\mathbf{x}^l)$ is the output predicted by the teacher model and $\mathbb{1}(\cdot)$ is the indicator function. A total of $\text{len} * N$ prototypes from labeled images are stored in our memory bank, where N denotes the number of classes and len denotes the predefined number of prototypes per class. The PMB is a first in, first out (FIFO) queue. At each iteration, prototypes of B_l images in a mini-batch are computed by (7) and stored in the memory bank.

Assuming that a given feature map $Z \in \mathbb{R}^{C\times H\times W}$ is generated from an unlabeled image by the student model, we aim to bound Z with features from the PMB and thus bring the student model with consistent feature distribution over the overall training data. Specifically, we compute the similarity relation between the pixel features $\{z_i \in Z : i \in (1, \ldots, H \times W)\}$ and the features in the PMB $\{\eta_j \in \Gamma : j \in (1, \ldots, \text{len}, \text{len}+1, \ldots, \text{len}*N)\}$ by the cosine similarity function $\text{sim}(u, v) = u^T v/\|u\|\|v\|$, which is suitable for computing the relationship between high-dimensional feature vectors. Meanwhile, a softmax layer is employed to obtain a normalized distribution of the pixel–prototype relation

$$q_i = \frac{\exp(\text{sim}(z_i, \eta_j)/t)}{\sum_{k=1}^{\text{len}*N} \exp(\text{sim}(z_i, \eta_k)/t)} \tag{8}$$

where $q_i \in Q, i \in (1, \ldots, H \times W)$ and $Q \in \mathbb{R}^{(\text{len}*N)\times H\times W}$. The parameter t represents the temperature coefficient for controlling the sharpness of the distribution.

Then, we aggregate q_i into N dimensional space by summing the pixel–prototype similarities that share the same class property of prototypes

$$q_i^{\text{agg}} = \sum_{k=1}^{\text{len}*N} \mathbb{1}[\text{class}(\eta_k) = i] \cdot q_k \tag{9}$$

where $q_i^{\text{agg}} \in Q^{\text{agg}}, i \in (1, \ldots, H \times W)$, and $Q^{\text{agg}} \in \mathbb{R}^{N\times H\times W}$.

Finally, in order to enforce class constraints on the calculated relation distribution, the Kullback–Leibler divergence is applied to compute the loss of Q^{agg} with soft logits of the

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pseudo-label $P^u \in \mathbb{R}^{N\times H\times W}$, which can be defined as

$$L_{\text{pro}} = KL(Q^{\text{agg}}||P^u) = \frac{1}{H\times W}\sum_{j=1}^{H\times W}\sum_{i=1}^{N} q_{i,j}^{\text{agg}} \log\left(\frac{q_{i,j}^{\text{agg}}}{p_{i,j}^u}\right) \tag{10}$$

where the soft logits of the pseudo-label P^u are generated by normalizing the intermediate output of the teacher's decoder before upsampling using the softmax function.

Under the explicit feature constraint of soft pseudo-label P^u, the feature descriptions of student model on the unlabeled images converge to the prototype representation on the labeled images. Furthermore, the PFC module improves the feature consistency over the whole training data with the unbalanced data distribution.

D. RCL Module

Training reliable pixel pseudo-labels allows to achieve entropy minimization on unlabeled samples while expanding the training samples. However, the model shows inferior generalization to pseudo-labels of unlabeled images for the classes with insufficient annotation in RSIs. Contrastive learning [51] usually uses unlabeled data for model pretraining, in which the samples are forced to have the same predictions as the augmented versions and differ from the predictions of the other samples. In some recent semisupervised methods, contrastive learning has been applied in various ways to train unlabled images, e.g., ReCo [27] and PC2Seg [28]. Inspired by them, we expect to impose additional constraints on the class attributes of unlabeled images so that features of the same class attract and different classes repel.

We suggest a graph-based RCL module in Fig. 7. Given an unlabeled image $\mathbf{x}^u$, the result with weak augmentation $A_w(\mathbf{x}^u)$ is fed to the teacher network and the result with strong augmentation $A_s(A_w(\mathbf{x}^u))$ is fed to the student network. Concretely, we use random flip, resize, and rotation for weak augmentation. Strong augmentation comprises a series of color jitters (brightness, contrast, saturation, and hue).

First, we assume that the intermediate output of the teacher for image $\mathbf{x}^u$ before upsampling is normalized using the softmax function to result in $P^u \in \mathbb{R}^{N\times H\times W}$, where the pixel predictions are $\{p_i\}_{i=1}^{H\times W}$. The pseudo-label graph is constructed with P^u to obtain a similarity matrix W^p with the size of $HW \times HW$, which is denoted as

$$W_{ij}^p = \begin{cases} 1, & \text{if } i = j \\ p_i \cdot p_j, & \text{if } i \neq j \text{ and } p_i \cdot p_j \geq T \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where T is a threshold parameter and the similarity of predictions between pairs of pixels below T is set to 0, which means that they are not connected to each other in the pseudo-label graph. Also, each pixel has the strongest connection to itself and the edge is set to 1.

Then, we utilize the feature output of the student's encoder on the strongly augmented image $Z = f_{\theta^-}(A_s(A_w(\mathbf{x}^u)))$ and the teacher's encoder on the weakly augmented image $Z' = f_{\xi^-}(A_w(\mathbf{x}^u))$ for the construction of pixel feature embedding graph. We build the pixel-embedding graph W^z as

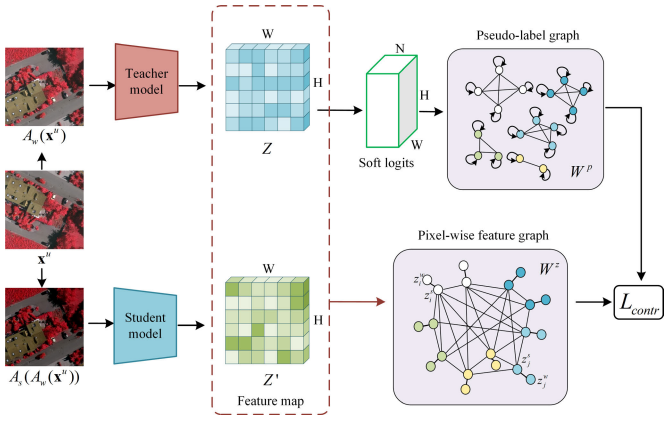


Fig. 7. Diagram of the RCL module. Strong data augmentation is injected into the student model. The constructed pseudo-labeled graph guides the student model to train consistent pixel features under perturbations and similar pixel features in similar classes.

$$W_{ij}^z = \begin{cases} z_i \cdot z_i', & \text{if } i = j \\ z_i \cdot z_j, & \text{if } i \neq j \end{cases} \tag{12}$$

where $z_i \in Z, i \in (1, \ldots, H \times W)$ and $z_i' \in Z', i \in (1, \ldots, H \times W)$.

The goal of our RCL module is to train the student model guided by the pseudo-label graph so that its feature pixel similarity graph has the same structure. Thus, we calculate the loss of the feature similarity graph with the pseudo-label graph by using the L_2 loss function, after normalizing W^z and W^p with $\hat{W}_{ij} = W_{ij}/\sum_j W_{ij}$. The RCL loss is formulated as follows:

$$L_{\text{contr}} = \frac{1}{HW \times HW}\sum_{i=1}^{HW}\sum_{j=1}^{HW}\left\|\hat{W}_{ij}^z - \hat{W}_{ij}^p\right\|_2^2. \tag{13}$$

The loss L_{contr} can be decomposed into two parts

$$\sum_{i=1}^{HW}\left\|\hat{W}_{ii}^z - \hat{W}_{ii}^p\right\|_2^2 + \sum_{i=1}^{HW}\sum_{j=1, j\neq i}^{HW}\left\|\hat{W}_{ij}^z - \hat{W}_{ij}^p\right\|_2^2. \tag{14}$$

The first term constrains the feature z_i to be similar to the feature z_i'. Concretely, it enforces the network to give consistent feature representations for the input images with different views, which results in invariant pixel features under perturbations for unlabeled images. The second term constrains the node-to-node similarity relationship in the graph structure. For the edge with W_{ii}^p more than T, it makes the features with the same predicted category more similar and conversely, the pixel features with different prediction classes are imposed away from each other.

E. Optimization Process

Based on the proposed label propagation and contrastive regularization strategy, our overall optimization process is described in Algorithm 1. To prevent unreliable pseudo-labels,

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Algorithm 1 Optimization Process of Our Method

Input: Labeled training set $\mathcal{D}_l = \left\{\left(\mathbf{x}_i^l, \mathbf{y}_i^l\right)\right\}_{i=1}^{N_l}$, unlabeled training set $\mathcal{D}_u = \left\{\mathbf{x}_i^u\right\}_{i=1}^{N_u}$, weak/strong augmentations A_w/A_s, and warm-up epoch $E_{warm-up}$.
Output: Fully trained student network f_θ.

Initialize the student network f_θ and teacher network f_ξ. Initialize the prototype memory bank with the length of $len * N$.
if $epoch < E_{warm-up}$ **then**
  for $\left\{(\mathbf{x}_k^l, \mathbf{y}_k^l)\right\}_{k=1}^{B} \subset \mathcal{D}_l$ **do**
    Train f_θ on $\mathcal{D}_l$ with cross-entropy loss L_{sup}.
    Calculate prototypes of $f_{\xi^-}(\mathbf{x}_k^l)$ and store them in the prototype memory bank.
    Update f_ξ with $\xi_t \leftarrow \tau\xi_{t-1} + (1-\tau)\theta_t$.
  end for
else
  for $\left\{(\mathbf{x}_k^l, \mathbf{y}_k^l)\right\}_{k=1}^{B} \subset \mathcal{D}_l$, $\left\{\mathbf{x}_k^u\right\}_{k=1}^{B} \subset \mathcal{D}_u$ **do**
    Obtain the pseudo-label $\hat{\mathbf{y}}_k^u = \arg\max f_\xi(\mathbf{x}_k^u)$.
    Calculate prototypes of $f_{\xi^-}(\mathbf{x}_k^l)$ and store them in the prototype memory bank.
    Random copy-paste on $(\mathbf{x}_k^u, \hat{\mathbf{y}}_k^u)$ with $(\mathbf{x}_k^l, \mathbf{y}_k^l)$.
    Employ the student network f_θ on unlabeled image $\mathbf{x}_k^u$: $\{Z_s\} \leftarrow f_\theta(A_s(A_w(\mathbf{x}_k^u)), \theta)$
    Employ the teacher network f_ξ on unlabeled image $\mathbf{x}_k^u$: $\{Z_w, P^u\} \leftarrow f_\xi(A_w(\mathbf{x}_k^u), \xi)$
    Calculate the L_{pro} (Eq. 10) through the PFC module.
    Calculate the L_{contr} (Eq. 13) through the RCL module.
    Calculate the L_{sup} (Eq. 2) on labeled image.
    Calculate the L_u (Eq. 15) on unlabeled image.
    Update f_θ with the overall objective function $loss$ (Eq. 17).
    Update f_ξ with $\xi_t \leftarrow \tau\xi_{t-1} + (1-\tau)\theta_t$.
  end for
end if
return f_θ

we follow the work [52] weighting the pixel loss by the sharpening pseudo-label confidence, and (3) is modified as

$$L_u = \frac{1}{|B_u|}\sum_{\mathbf{x}_i^u \in B_u}\frac{1}{H'W'}\sum_{j=1}^{H'W'} w_{ij}\ell_{ce}\left(f_\theta\left(\mathbf{x}_i^u;\theta\right)_j, \hat{\mathbf{y}}_{ij}^u\right) \quad (15)$$

$$w_{ij} = \max\left(f_\xi\left(\mathbf{x}_i^u\right)_j\right)^\gamma \quad (16)$$

where γ is set to 6. We set the warm-up epoch $E_{\text{warm-up}}$ and calculate only the supervised loss during the warm-up epoch. The overall objective function is defined as

$$\text{loss} = L_{\text{sup}} + L_u + \lambda_1 L_{\text{pro}} + \lambda_2 L_{\text{contr}} \quad (17)$$

where λ_1 and λ_2 are loss weights to controls their contribution.

IV. Experiments

We comprehensively evaluate the proposed framework in this section, including the public ISPRS semantic labeling benchmark,[1] and the AIR-SEG dataset [9] that provides additional unlabeled images. Datasets and implementation details are presented. We report an extensive ablation study on the ISPRS Vaihingen dataset and comparative results with other methods.

A. Datasets

1) ISPRS Vaihingen: The Vaihingen dataset provides 33 true orthophoto (TOP) tiles and the corresponding semantic labels. The size of these images ranges from 1996 × 1995 pixels to 3816 × 2550 pixels, and the images are composed of near-infrared, red, and green bands with a spatial resolution of 9 cm. There are six classes labeled in the dataset, including five foreground classes and one background class. In our experiments, 16 images are used for training, and the remaining 17 are used for testing. We follow the partitioning protocol of many semisupervised works [32], [50] and divide the entire training set into two groups for evaluation by randomly selecting 1/16, 1/8, and 1/4 of the training set as the labeled set and the rest of the images as the unlabeled set.

2) ISPRS Potsdam: The Potsdam dataset contains 38 high-resolution images and the corresponding semantic labels with a size of 6000 × 6000 pixels. It has the same spatial resolution and category annotation as the Vaihingen dataset. We conduct experiments with images composed of red, green, and blue bands, comprising 24 training images and 14 test images. We compare our framework with other semisupervised methods under 1/12, 1/8, and 1/4 partition protocols.

3) AIR-SEG: The AIR-SEG is a challenging semantic segmentation dataset of RSIs published by Sun et al. [9]. These satellite images are collected from Google Earth, covering different areas of South China. The images include red, green, and blue bands with a spatial resolution of 0.27 m and a 2000 × 2000 pixels size. This dataset includes nine foreground object classes and one background class. There are 29 images for training and 28 images for testing. Also, 72 unlabeled images are provided as additional training data. Thus, we verify the improvement of our semisupervised approach to fully supervised segmentation models by effectively utilizing a large number of unlabeled images in Section IV-E.

B. Implementations

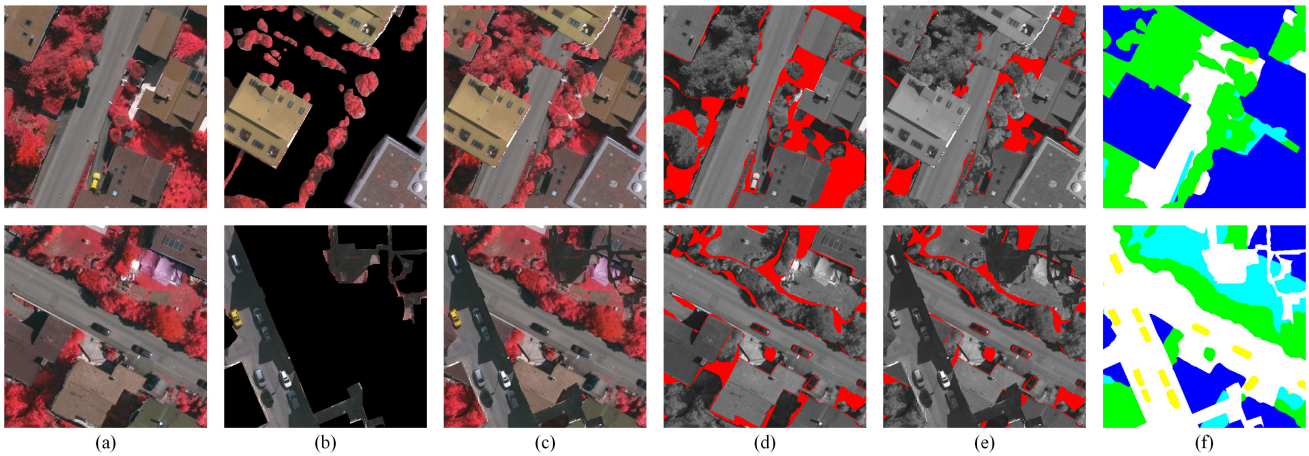
1) Training Setting: We compare our method with other methods using different segmentation models, including Deeplabv3+ [53], Deeplabv3 [39], and HMANet [6]. They all employ ResNet-50 [54] pretrained on the ImageNet [55] as the backbone and apply dilated convolutions in the last two stages to keep the output stride as 8. We use the stochastic gradient descent (SGD) strategy for parameter optimization of models with an initial learning rate of 0.01. The momentum is set to 0.9, and the weight decay is 0.0001. The poly scheduling [56] is used to decay the learning rate by $\text{lr}_{\text{base}} \cdot (1 - (\text{iter}/\text{total iter}))^{0.9}$. We follow the training rules in [57] to ensure a fair comparison. Our networks are trained for 40 epochs with a batch of eight images. For the fully supervised baseline, there are eight labeled images in a mini-batch.

[1] http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html

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TABLE I
EXPERIMENTAL RESULTS OF THE PROPOSED DIFFERENT MODULES ON THE VAIHINGEN DATASET

Method	1/8							1/4						
	F1 score per category					OA	mIoU	F1 score per category					OA	mIoU
	Imp. surf.	Build.	Low veg.	Tree	Car			Imp. surf.	Build.	Low veg.	Tree	Car		
Suponly	87.68	90.38	73.96	77.77	77.05	82.65	69.10	88.11	91.93	75.95	83.24	78.74	84.99	72.25
Adaptive L_u	88.78	92.04	78.75	81.33	79.36	85.26	72.87	89.14	92.33	80.75	84.97	81.45	86.82	75.29
+SDA	88.78	92.50	78.61	81.23	80.00	85.38	73.14	89.31	93.29	80.74	85.86	81.03	87.30	75.83
+RCP	88.92	92.16	78.96	80.86	80.90	85.32	73.31	89.57	93.27	80.99	85.90	81.53	87.43	76.13
+SDA+RCP	88.89	93.10	79.04	81.45	80.99	85.59	73.84	89.44	93.51	80.89	86.02	81.94	87.44	76.30
+L_{pro}	88.79	92.43	79.53	82.69	79.47	85.82	73.64	89.40	93.25	81.24	85.85	81.93	87.42	76.24
+SDA+L_{pro}	89.01	92.52	79.34	82.91	79.96	85.92	73.90	89.61	93.31	81.05	86.11	82.11	87.54	76.41
+SDA+L_{contr}	89.08	92.33	78.77	81.24	80.12	85.45	73.25	89.48	93.44	80.56	85.75	82.44	87.32	76.25
+RCP+L_{pro}+SDA+L_{contr}	89.46	93.24	79.37	81.97	80.81	**86.09**	**74.26**	89.70	93.60	81.25	86.09	82.01	**87.65**	**76.56**



(a) (b) (c) (d) (e) (f)

Fig. 8. Visual analysis about RCP on the Vaihingen dataset. (a) Unlabeled images. (b) Selected classes in labeled images. (c) Unlabeled images after RCP. (d) Grayscale images of (a). (e) Grayscale image of (c). (f) Pseudo-labels of unlabeled images after RCP. The red pixels in (d) and (e) indicate predictions of wrong classification in the pseudo-labels. Legend—white: impervious surfaces, blue: building, cyan: low vegetation, green: tree, and yellow: car.

For semisupervised training, a batch consists of four labeled images and four unlabeled images, and one epoch refers to the entire iteration of all unlabeled datasets. The labeled subset is repeated several times inside an epoch. The warm-up epoch is set empirically to 6 following [45]. The original image is sliced into 512 × 512 pixels for training and testing.

2) Evaluation Metrics: We test cropped images on single scale, applying evaluation metrics, including overall accuracy (OA), mean of intersection over union (mIoU), and F1 score, which are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{18}$$

$$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C} \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{19}$$

$$\mathrm{F}_1 = (1+\beta^2) \cdot \frac{\mathrm{presion} \cdot \mathrm{recall}}{\beta^2 \cdot \mathrm{presion} + \mathrm{recall}}, \quad \beta = 1 \tag{20}$$

in which TP, TN, FP, and FN are the numbers of true positives, true negatives, false positives, and false negatives, respectively, and C indicates the number of classes. We report the OA metric of models for all classes and the mIoU and mF1 metrics are calculated for foreground classes.

C. Experiments on Vaihingen Dataset

We first conduct ablation studies on the Vaihingen dataset with 1/8 and 1/4 of the training set labeled and analyze the effect of each component. Then, we compare with state-of-the-art semisupervised methods and discuss the versatility of our framework.

1) Ablation Experiments: We conduct experiments by adding different modules successively with Deeplabv3+. Table I reports the evaluation results for each group of our experiments on the test set. Suponly denotes without using any unlabeled data. Adaptive L_u refers to training unlabeled data using (15). SDA denotes that the model is trained with adaptive L_u using the strong data augmentation mentioned in Section III-D. From Table I, the mIoU of the models trained with supervision only is 69.10% and 72.25% for the 1/8 and 1/4 protocols, respectively. The baseline with adaptive L_u improves the mIoU metrics of the model to 72.87% and 75.29%. The SDA is applied to the student model to bring an improvement of 0.27% and 0.54%, indicating the effectiveness of FixMatch [22] on semantic segmentation.

a) Effect of RCP: The proposed RCP is applied as a data augmentation method between the labeled and unlabeled images. As can be seen in Table I, our RCP enhances 0.44%

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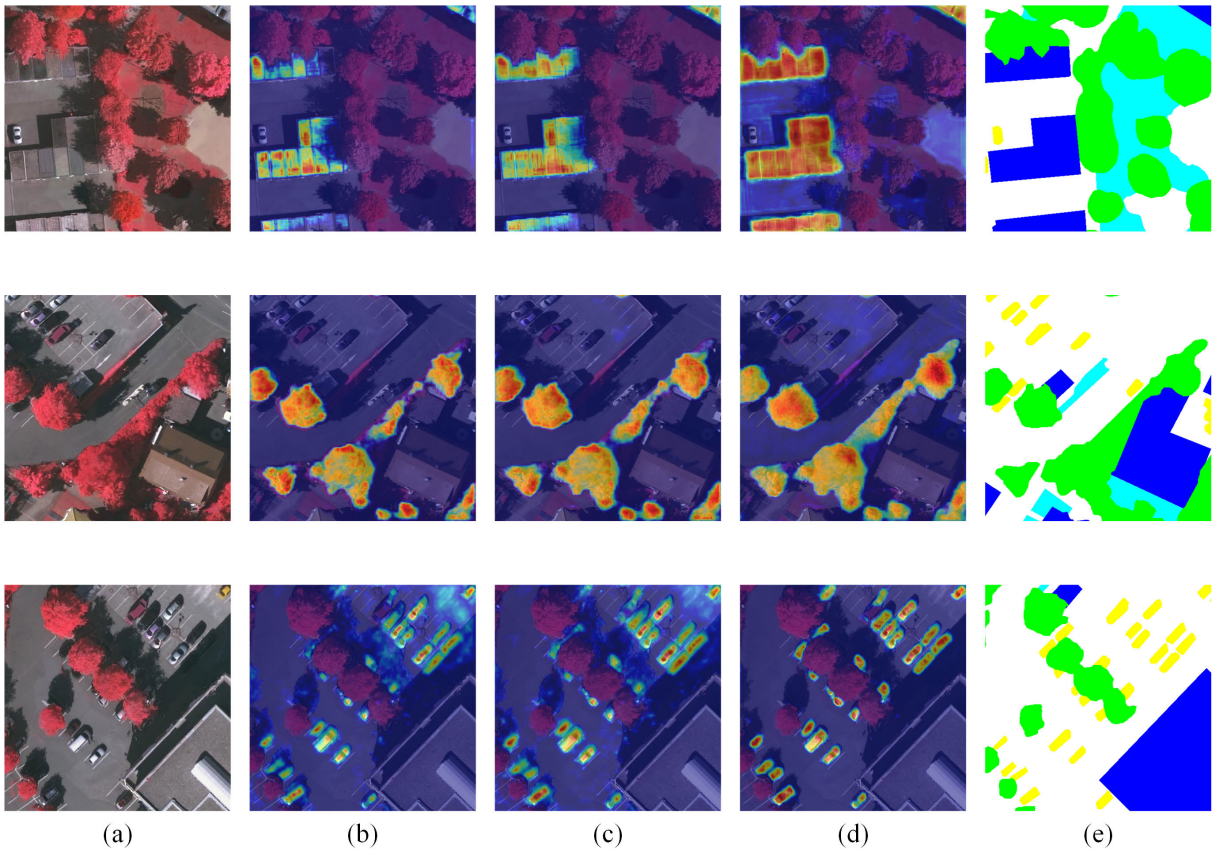


Fig. 9. Grad-CAM visualization of the feature response maps on unlabeled images on the Vaihingen dataset with 1/4 labeled data. (a) Unlabeled images. (b) Supervision only. (c) With adaptive L_u. (d) With the proposed PFC module. (e) Ground truth of unlabeled images. The first row indicates building, the second row indicates tree, and the last row indicates car.

and 0.84% mIoU comparing with the adaptive L_u baseline under 1/8 and 1/4 protocols, respectively. Also, jointly with SDA and RCP, the model performance is further improved. We analyze the effectiveness of the RCP in Fig. 8. Fig. 8(a) shows the unlabeled images used for training. Fig. 8(b) shows the classes extracted from the labeled images by random selection, e.g., classes in the first row belong to building and tree, and classes in the second row belong to Imper. surf. and car. Fig. 8(c) shows the augmented unlabeled images after our RCP. For better observation, Fig. 8(d) and (e) shows the grayscale images of Fig. 8(a) and (c), respectively, where the red pixels indicate predictions of wrong classification in the pseudo-labels. Fig. 8(f) shows pseudo-labels for the unlabeled images augmented by the RCP. It can be observed that in the first row, the mispredicted pixels are replaced with pixels from the annotated image. Thus, the model is prevented from learning the wrong pseudo-label partly. Besides, in the second row, the original pseudo-label is difficult to produce accurate segmentation at the edges of objects, such as car. With the RCP, unlabeled images are fused with Impervious surfaces and car, which have clear semantic boundaries. Thus, our RCP prompts the model to learn favorable pseudo-labels.

b) Effect of PFC module: With feature constraints executed on the unlabeled data, our PFC module achieves 0.77% and 0.98% mIoU improvement based on adaptive L_u in the 1/8 and 1/4 protocols, respectively. In particular, for tree and low vegetation with unbalanced data distribution under the 1/8 protocol, our method L_{pro} improves by 1.36% and 0.78%, respectively. Also, the model performance is increased by combining SDA and PFC module.

TABLE II
COMPARISON RESULTS USING HARD LABEL OR SOFT LABEL OF THE PFC MODULE ON THE VAIHINGEN DATASET

Model	Method	OA(%)	mIoU(%)	mF1(%)
Deeplabv3+	Hard Lable	85.80	73.26	84.32
with 1/8 labeled data	Soft Lable	**85.82**	**73.64**	**84.58**
Deeplabv3+	Hard Lable	87.30	76.02	86.19
with 1/4 labeled data	Soft Lable	**87.42**	**76.24**	**86.33**

Fig. 9 shows the feature improvement of our model for unlabeled images. We use a general feature visualization tool Grad-CAM [58] to represent the class capture capability of different models for unlabeled images. Specifically, the model with supervision only has not a complete response for capturing the building, tree, and car. Training with adaptive L_u boosts the generalizability of the model to different scenes, causing the category feature activation to be enhanced. The model leveraging our PFC module can capture more accurate and complete category regions of building, tree, and car.

The soft pseudo-labels from teacher are employed in (10) to calculate the constraint loss L_{pro} on unlabeled data. In Table II,

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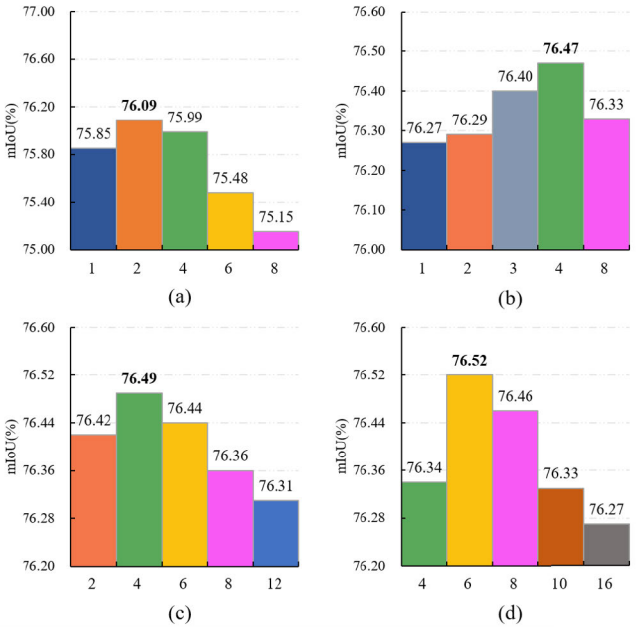


Fig. 10. Effect of len of the PMB with the different batch size settings. (a) Batch size is set to 2. (b) Batch size is set to 4. (c) Batch size is set to 6. (d) Batch size is set to 8.

TABLE III

EFFECT OF THE TEMPERATURE COEFFICIENT IN THE PFC MODULE ON THE VAIHINGEN DATASET WITH 1/4 LABELED DATA

t	0.07	**0.1**	0.3	0.5	0.7
OA(%)	87.54	**87.63**	87.54	87.51	87.53
mIoU(%)	76.43	**76.47**	76.37	76.30	76.26
mF1(%)	86.45	**86.46**	86.41	86.36	86.33

TABLE IV

EFFECT OF THE THRESHOLD IN THE RCL MODULE ON THE VAIHINGEN DATASET WITH 1/4 LABELED DATA

T	mIoU(%)	OA(%)	mF1(%)
0.1	75.95	87.24	86.12
0.2	76.10	87.30	86.23
0.3	**76.25**	**87.32**	**86.33**
0.4	76.00	87.29	86.16
0.5	75.95	87.21	86.13

we compare different constraint methods, including soft labels and hard labels. Hard labels with a one-hot format force the features to close to the class with the highest prediction probability, while soft labels contain the correlations of the classes in the prediction. It can be seen that soft pseudo-labels achieve higher accuracy. The result can be attributed to the fact that hard labels may introduce incorrect class constraints, and soft labels are more applicable to the multicategorical feature constraint.

We compare and analyze the effect of the number of prototype len in the PMB. Based on the 1/4 protocol of the Vaihingen dataset, Fig. 10(a)–(d) shows the experimental results when the batch size is set to 2, 4, 6 and 8, respectively. It can be seen that as the number of prototypes len increases, the mIoU metric presents an increase and then a decrease. For the experiments with a batch size of 2 and 4, len for every class set to 2 and 4 brings the highest accuracy gain, respectively. We consider that since there are two or four labeled images in a batch for training, and the earlier prototype features are obsolete with the update of the teacher model's weights that brings negative effects. For the experiments with a batch size of 6 and 8, the highest accuracy is achieved when len is equal to 4 and 6, respectively. We can notice that compared to the results in Fig. 10(a) and (b), the increase in the number of prototype features used peaks earlier rather than at equal to the batch size. Therefore, in addition to the impact of the update of the model weights, more prototype features could bring noise and thus have a negative impact. Besides, we evaluate the parameter t in (8), which controls the sharpness of the similarity distribution. From Table III, the highest accuracy is obtained when $t = 0.1$.

c) Effect of RCL module: Our RCL module uses class correlation to guide the separability and the class consistency of features on unlabeled images. From Table I, the RCL module attains 0.11% and 0.42% mIoU boost over the 1/4 and 1/8 protocols based on the SAD, respectively. We analyze the effectiveness of the RCL module in Fig. 11, which shows the segmentation results and corresponding t-distributed stochastic neighbor embedding (t-SNE) [30] visualization of features on the test images. It can be observed that the Suponly model produces worse segmentations and confusion between different classes of features. The addition of unlabeled data for training leads to enhanced performance of the model in Fig. 11(b) and (c). By comparison, our RCL module effectively improves the segmentation results and clarifies the decision boundaries of features.

In addition, we test the effect of the threshold T in (11). As shown in Table IV, the RCL module acquires the highest effectiveness when T is set to 0.3.

d) Combination: Overall, in Table I, with our overall framework, the mIoU metric of models improves from 69.10% and 72.25% to 74.26% and 76.56% under 1/8 and 1/4 partition protocols, obtaining an improvement of 5.16% and 4.31%, respectively. In particular, the F1 scores of low vegetation, tree, and car under 1/8 protocol increase by 5.41%, 4.20%, and 3.76%, respectively.

e) Ablation study on hyperparameters: We analyze the impact of various hyperparameters of our method on the Vaihingen and Potsdam datasets with the 1/4 partition protocol and the AIR-SEG dataset. Fig. 12 shows the impact for the EMA control parameter τ in (1) and two balance parameters of the loss function λ_1 and λ_2 in (17). The parameter τ is used to control the degree of updating of the teacher model through the EMA. In general, the smaller the value of τ, the greater the update magnitude of the teacher model's weights. According to our experimental results, we can see that $\tau = 0.999$ brings more stable updates for the teacher model on all the three datasets. Moreover, we analyze the sensitivity of two balance parameters of the loss function. To balance the contributions of different loss functions, we adjust the loss weights so

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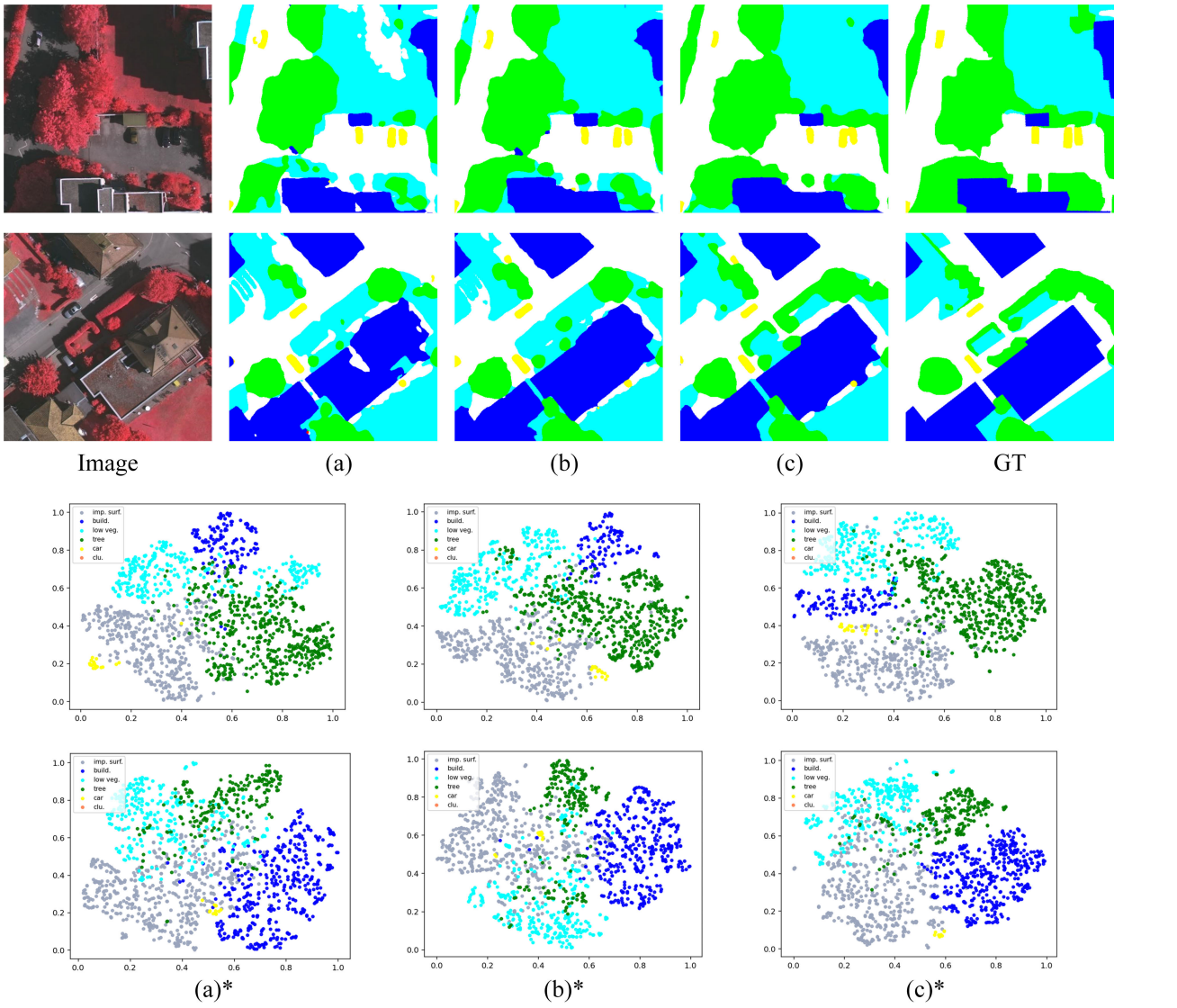


Fig. 11. Segmentation results and t-SNE visualization of feature spaces on test images of Vaihingen dataset with 1/4 labeled data. (a) Supervision only. (b) With Adaptive L_u. (c) With the proposed RCL module. * denotes the corresponding t-SNE visualization generated by different methods.

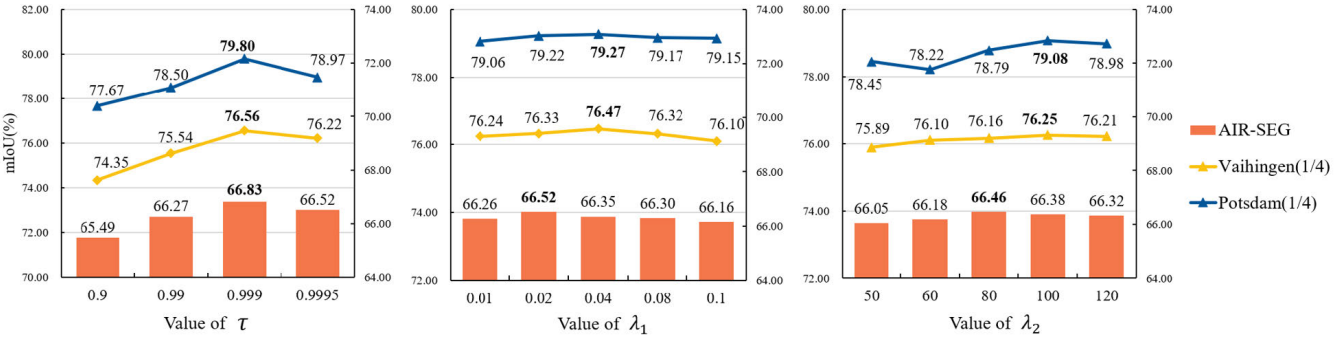


Fig. 12. Ablation study of hyperparameters on different datasets. τ is the control parameter of EMA and λ_1 and λ_2 are the loss weights.

that the loss values are in the same order of magnitude as the unsupervised loss L_u empirically. Our method achieves the highest accuracy results at $\lambda_1 = 0.04$ and $\lambda_2 = 100$ for the Vaihingen and Potsdam datasets and $\lambda_1 = 0.02$ and $\lambda_2 = 80$ for the AIR-SEG dataset. Overall, our method is insensitive to the value of loss weight parameters on different datasets, and the PFC and RCL loss weights are lower on the AIR-SEG dataset that contains more categories.

2) Comparison With State-of-the-Art Semisupervised Methods: Semisupervised methods used for comparison consist of MT [31], ST [20], ClassMix [25], AEL [26], and U2PL [45]. Table V shows the comparison results under 1/16, 1/8,

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TABLE V
COMPARISON WITH SEMISUPERVISED METHODS ON THE VAIHINGEN DATASET

Labeled Data	Method	F1 score per category(%)					mF1(%)	OA(%)	mIoU(%)
		Imper. surf.	Building	Low veg.	Tree	Car			
1/16	Suponly	82.59±2.47	83.02±3.06	65.58±3.01	79.38±1.95	61.81±8.85	74.48±1.93	77.83±1.07	60.29±2.26
	MT [31]	85.82±0.51	88.16±0.47	68.95±1.23	80.37±1.29	64.98±5.58	77.66±1.27	81.08±0.34	64.43±1.38
	ST [20]	85.76±0.96	87.94±1.81	66.92±1.77	79.41±2.25	63.79±2.83	76.76±1.08	80.41±1.72	63.33±1.54
	ClassMix [25]	86.60±0.33	89.26±1.00	68.92±1.66	78.77±1.17	69.11±4.30	78.53±1.28	80.94±0.97	65.50±1.59
	AEL [26]	87.52±0.25	90.73±0.84	68.92±1.90	78.75±1.48	73.54±3.42	79.89±1.11	81.79±0.51	67.33±1.42
	U2PL [45]	**87.70**±0.23	**91.34**±0.96	67.09±3.04	77.90±1.56	72.66±3.01	79.34±1.48	81.76±0.56	66.84±1.70
	Ours	87.19±0.61	90.31±1.27	**71.00**±3.38	**81.14**±1.54	**75.29**±3.62	**80.99**±1.51	**82.73**±0.82	**68.71**±1.90
1/8	Suponly	84.93±2.20	87.93±2.68	73.93±1.17	82.07±3.05	68.43±6.92	79.46±1.64	82.04±0.69	66.61±2.12
	MT [31]	87.00±0.89	90.58±1.29	75.75±3.06	82.62±1.73	74.99±2.82	82.19±1.32	83.94±1.26	70.26±1.83
	ST [20]	86.83±1.14	90.92±0.51	74.85±1.24	81.87±1.83	75.62±3.42	82.02±0.91	83.55±0.51	70.03±1.25
	ClassMix [25]	86.63±1.73	90.55±2.26	77.66±1.82	83.74±2.47	76.77±3.23	83.07±0.89	84.43±0.54	71.44±1.30
	AEL [26]	87.78±1.27	**92.25**±0.97	76.12±2.41	82.34±1.13	77.11±3.67	83.12±1.41	84.55±0.98	71.64±1.99
	U2PL [45]	87.59±0.92	91.22±1.09	78.31±1.12	**84.94**±0.80	75.98±2.84	83.61±0.81	85.43±0.57	72.34±1.35
	Ours	**88.30**±1.04	92.19±1.19	**78.57**±1.05	82.77±2.25	**79.29**±2.18	**84.22**±0.73	**85.45**±0.74	**73.14**±1.07
1/4	Suponly	87.69±0.33	91.09±0.59	76.90±0.88	84.23±0.78	76.71±1.41	83.33±0.21	85.01±0.28	71.85±1.01
	MT [31]	88.68±0.32	91.96±0.29	78.62±0.81	84.96±0.37	79.22±0.67	84.69±0.17	86.18±0.29	73.80±0.27
	ST [20]	88.74±0.40	92.74±0.38	79.17±0.68	84.92±0.29	80.22±0.74	85.16±0.25	86.39±0.42	74.50±0.40
	ClassMix [25]	89.16±0.20	92.47±0.31	79.54±0.79	85.54±0.27	80.24±0.66	85.39±0.27	86.75±0.12	74.84±0.38
	AEL [26]	88.93±0.26	92.74±0.53	80.14±0.57	85.15±0.92	80.33±0.15	85.46±0.16	86.70±0.27	74.93±0.25
	U2PL [45]	89.30±0.22	93.32±0.20	80.76±0.18	85.52±0.44	81.45±0.44	86.07±0.27	87.19±0.25	75.86±0.41
	Ours	**89.57**±0.16	**93.47**±0.17	**81.01**±0.17	**85.93**±0.20	**81.94**±0.21	**86.39**±0.15	**87.48**±0.14	**76.34**±0.24

TABLE VI
COMPARISON WITH SEMISUPERVISED METHODS ON THE VAIHINGEN DATASET USING DIFFERENT MODELS

Network	Method	F1 score per category(%)					mF1(%)	OA(%)	mIoU(%)
		Imper. surf.	Building	Low veg.	Tree	Car			
Deeplabv3 [39]	Suponly	88.39	91.03	74.79	73.89	77.29	81.08	82.46	68.81
	ClassMix [25]	89.20	**92.64**	77.82	78.31	80.46	83.69	84.75	72.43
	U2PL [45]	88.80	91.61	77.91	81.12	79.77	83.84	85.67	72.87
	Ours	**89.56**	92.51	**79.05**	**81.52**	**80.91**	**84.71**	**85.77**	**73.85**
HMANet [6]	Suponly	88.34	91.25	76.19	78.09	77.38	82.25	83.68	70.35
	ClassMix [25]	89.47	92.63	79.03	81.19	80.97	84.66	85.71	73.78
	U2PL [45]	88.89	**93.10**	79.04	81.45	**80.99**	84.69	85.59	73.84
	Ours	**89.48**	92.30	**79.63**	**83.34**	80.52	**85.05**	**86.23**	**74.33**

and 1/4 protocols for the Vaihingen dataset. We reproduce the compared methods on Deeplabv3+ with a backbone of ResNet50. Considering the randomness of sampling, we repeat the experiment ten times and report the mean and standard deviation of the model on the test data. Our approach substantially improves the model trained by supervision only, especially in low vegetation and car. Compared with other semisupervised methods, our framework outperforms them under 1/16, 1/8, and 1/4 partition protocols. In particular, our approach outperforms the SOTA methods by 1.38%, 0.8%, and 0.48% mIoU under 1/16, 1/8, and 1/4 protocols, respectively. It can be seen that the Suponly model performs with large standard deviations by random sampling, and our method shows decreased standard deviations, verifying the robustness of the method for different data distributions, especially under the 1/8 and 1/4 protocols.

3) Effectiveness With Different Models: We carry out comparison experiments with different segmentation networks, including Deeplabv3 [39] and HMANet [6], to demonstrate the generality of our semisupervised approach with respect to the model. They both use ResNet50 as the backbone network, and HMANet is proposed for remote sensing scenes and has superior segmentation performance. Table VI shows their comparison results under the 1/8 protocol on the Vaihingen dataset. Our method results in 5.04% and 3.98% improvement in mIoU with only supervised training for deeplabv3 and HMANet, respectively. Moreover, our method outperforms U2PL on two networks, especially achieving higher F1 scores for low vegetation and tree. The experimental results demonstrate that our method still exhibits significant performance when used with different decoders.

D. Experiments on Potsdam Dataset

1) Comparison With State-of-the-Art Semisupervised Methods: Our method is further implemented on the ISPRS Potsdam dataset to proof its benefits. We randomly select

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TABLE VII
COMPARISON WITH SEMISUPERVISED METHODS ON THE POTSDAM DATASET

Labeled Data	Method	F1 score per category(%)					mF1(%)	OA(%)	mIoU(%)
		Imper. surf.	Building	Low veg.	Tree	Car			
1/12	Suponly	84.47±1.98	90.29±1.81	76.43±2.35	76.79±3.49	87.05±1.11	83.01±1.97	80.96±2.08	71.38±2.82
	MT [31]	85.96±0.93	91.19±2.31	76.96±3.16	78.99±1.71	87.33±1.74	84.09±1.03	82.27±1.14	72.95±1.43
	ST [20]	86.49±0.93	92.14±1.25	77.88±1.88	78.13±1.74	86.76±0.46	84.28±1.07	82.53±1.42	73.24±1.56
	ClassMix [25]	86.73±2.01	92.49±1.73	78.77±0.43	80.44±1.92	87.91±1.27	85.27±1.08	83.34±1.03	74.68±1.71
	AEL [26]	87.98±1.11	92.98±0.37	**79.34**±1.56	80.70±0.72	**88.63**±0.94	85.92±0.67	84.35±0.84	75.68±1.03
	U2PL [45]	87.52±0.64	92.89±0.41	78.85±1.52	80.36±3.22	88.25±0.77	85.57±0.99	83.87±1.26	75.17±1.40
	Ours	**88.02**±0.67	**93.46**±0.62	79.13±0.33	**81.58**±2.11	88.53±1.15	**86.14**±0.86	**84.36**±0.50	**76.03**±1.33
1/8	Suponly	86.81±1.15	91.37±1.69	78.04±1.82	78.92±2.91	87.44±1.24	84.51±1.35	82.94±1.54	73.56±1.98
	MT [31]	88.03±0.44	92.84±0.55	80.01±0.63	80.96±2.08	88.63±1.13	86.09±0.45	84.49±0.50	75.92±0.66
	ST [20]	87.50±0.97	92.45±0.76	79.79±0.73	81.42±0.59	88.49±0.56	85.93±0.56	84.39±0.64	75.63±0.88
	ClassMix [25]	88.33±1.12	92.88±0.62	80.00±0.78	80.71±1.66	89.13±0.41	86.21±0.37	84.59±0.64	76.11±0.56
	AEL [26]	87.51±0.58	92.84±0.23	80.18±0.71	82.06±0.35	88.94±0.22	86.30±0.42	84.76±0.40	76.20±0.64
	U2PL [45]	88.16±0.94	92.99±0.49	80.71±0.92	82.10±0.35	89.21±0.42	86.63±0.61	85.16±0.65	76.71±0.95
	Ours	**88.65**±0.63	**93.31**±0.27	**81.30**±0.54	**82.59**±0.37	**89.49**±0.35	**87.07**±0.41	**85.57**±0.46	**77.33**±0.59
1/4	Suponly	87.86±0.67	91.95±2.24	79.45±2.20	81.77±2.46	88.98±0.72	86.00±1.15	84.30±1.22	75.76±1.76
	MT [31]	88.93±0.61	93.86±0.73	80.79±1.03	81.59±1.76	89.71±0.60	86.97±0.79	85.44±0.86	77.30±1.20
	ST [20]	89.35±0.58	93.92±1.00	80.98±1.07	81.82±1.94	89.52±0.53	87.12±0.81	85.65±0.93	77.53±1.23
	ClassMix [25]	89.40±0.25	94.21±0.71	81.49±1.65	82.74±1.62	89.72±0.67	87.51±0.74	86.04±0.79	78.13±1.14
	AEL [26]	89.63±0.44	94.12±0.17	82.27±0.49	83.39±1.09	89.47±1.19	87.78±0.59	86.34±0.57	78.49±0.93
	U2PL [45]	89.81±0.23	94.27±0.45	82.07±0.53	83.53±0.77	89.90±0.97	87.92±0.45	86.55±0.35	78.73±0.71
	Ours	**90.00**±0.21	**94.82**±0.24	**83.19**±0.69	**84.24**±0.49	**90.26**±0.30	**88.50**±0.34	**87.21**±0.31	**79.64**±0.52

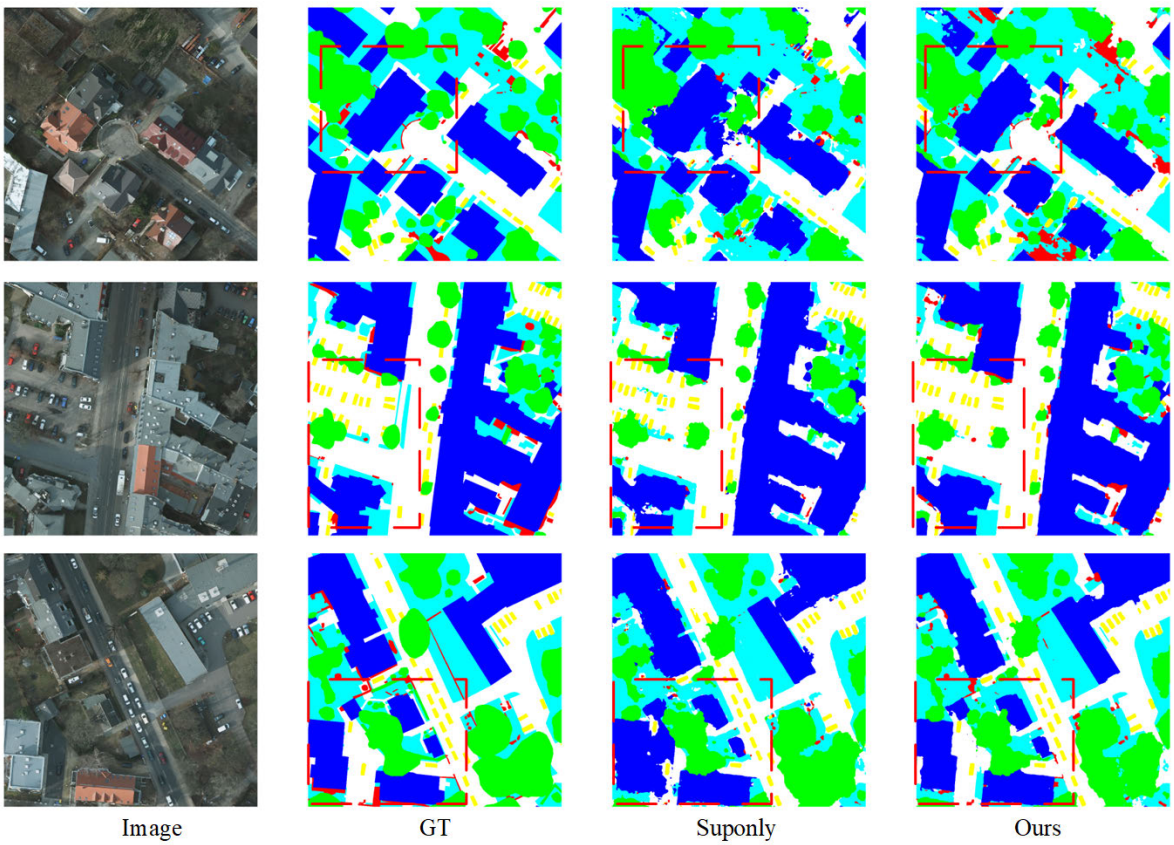


Fig. 13. Visualization results of our method on the Potsdam dataset. Legend—white: impervious surfaces, blue: building, cyan: low vegetation, green: tree, yellow: car, and red: clutter.

1/12, 1/8, and 1/4 of the training images as the labeled set and the rest as the unlabeled set. The results are given in Table VII, which reports the mean and standard deviation of ten random sampling experiments. Compared to the model trained with supervision only, our method achieves 4.65%, 3.77%, and 3.88% improvement of mIoU under 1/12, 1/8,

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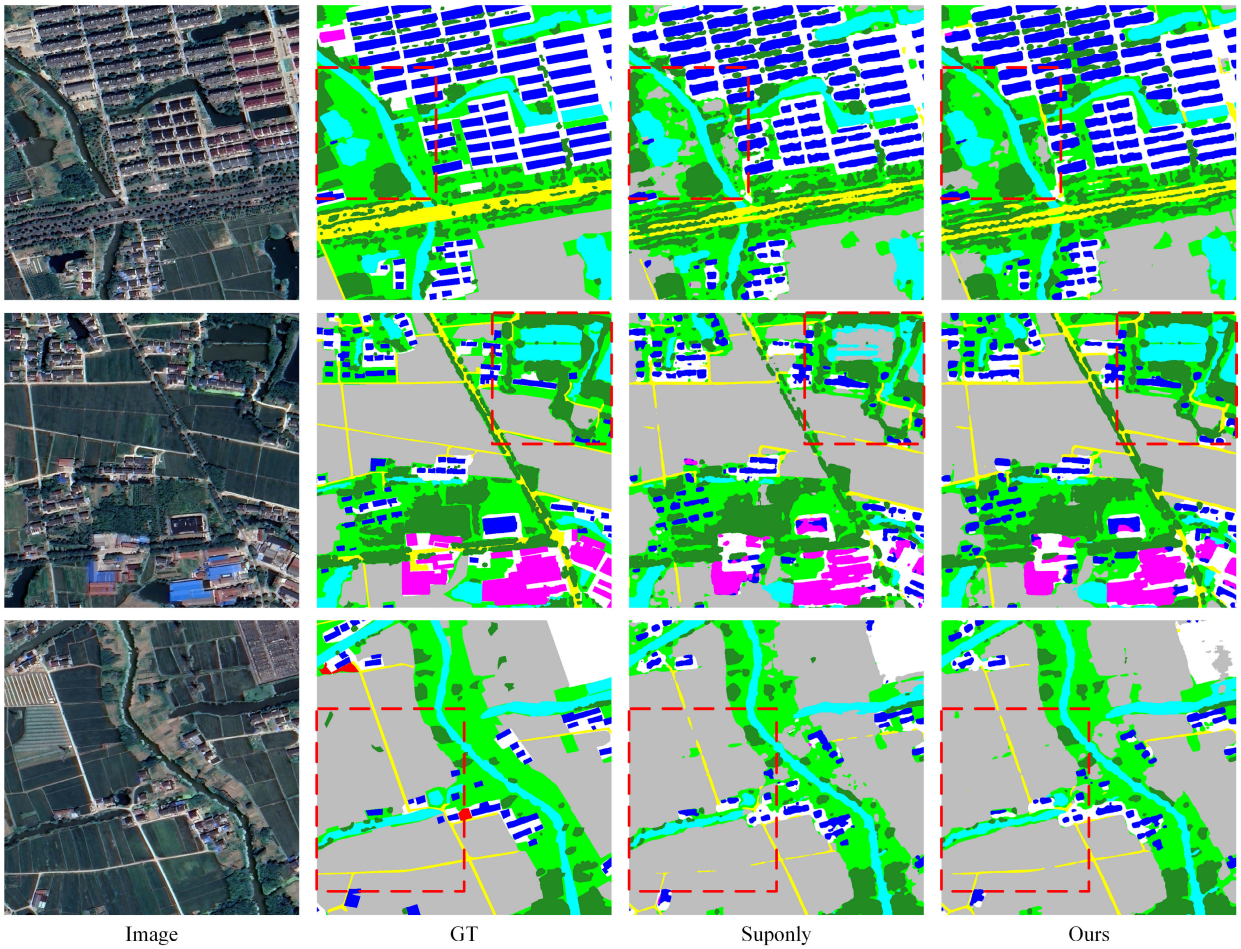


Fig. 14. Visualization results of our method on the AIR-SEG dataset. Legend—white: impervious surfaces, magenta: factory, blue: residential building, yellow: road, cyan: water, gray: farmland, red: bare land, green: grass, and forest green: tree.

and 1/4 protocols, respectively. Especially under 1/8 protocol, our method boosts 3.26% and 3.67% for low vegetation and tree, respectively. We perform a visual analysis in Fig. 13, showing the comparison for the Suponly and our method. Our method significantly improves the segmentation results compared to the model trained only with supervision, which captures objects' boundaries more accurately.

In addition, the average overall performance of our method surpasses other semisupervised methods under 1/12, 1/8, and 1/4 partition protocols in Table VII. This validates that our approach can overcome cases of unbalanced class distribution to enhance model generalization from unlabeled images more efficiently.

2) Time Cost of Training and Testing: We also analyze the time cost of different methods used for training and testing. Our experiments are all conducted on a NVIDIA GeForce RTX 3090. In Table VIII, we report the time cost for the training under 1/4 protocol and the testing of the test set on the Potsdam dataset. To train the unlabeled data, ST generates pseudo-labels using a pretrained model on the labeled images and retrains them. ST has the same computational complexity as the Suponly model in a single iteration, and its time cost is related to multistage training and pseudo-label generation. MT can simultaneously train labeled and unlabeled data from scratch and add consistency loss based on the segmentation loss. SSL methods that combine CR and ST, such as ClassMix, AEL, U2PL, and ours, are all modifications of the MT method. The design of additional data augmentations, modules, or optimizing objectives increases the time complexity of training. Compared with other methods, our method achieves the satisfactory performance while sacrificing training time. In addition, the test time is independent of the semisupervised strategy, and the average inference time of Deeplabv3+ on the test set is 1.43 min for ten times.

TABLE VIII
TIME COST OF TRAINING AND TEST TIME FOR DIFFERENT METHODS

Model	Method	Training	Test	mIoU(%)
Deeplabv3+ with 1/4 labeled data	Suponly	1.51h	1.43min	75.76
	MT [31]	21.3h		77.54
	ST [20]	12.5h		77.59
	ClassMix [25]	24.7h		78.37
	AEL [26]	25.9h		78.44
	U2PL [45]	23.1h		79.45
	Ours	29.4h		79.80

E. Experiments on AIR-SEG Dataset

The challenging AIR-SEG dataset is also experimented to show the generalization of our method. We use the entire training set (29 images) and additional 72 unlabeled images.

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TABLE IX
COMPARISON WITH SEMISUPERVISED METHODS ON THE AIR-SEG DATASET

Network	Method	IoU score per category(%)*									mIoU(%)	OA(%)
		IS	RB	Factory	Road	Water	Farmland	Grass	Tree	BL		
Deeplabv3 [39]	Suponly	59.38	64.42	70.39	71.65	83.66	79.76	53.05	80.61	25.10	65.34	82.15
	MT [31]	58.77	64.36	68.98	71.22	83.03	80.71	54.10	80.83	27.83	65.54	82.25
	ClassMix [25]	59.43	**65.51**	69.84	**72.64**	83.80	80.69	54.66	80.99	23.23	65.64	82.55
	AEL [26]	58.84	64.18	70.53	72.34	85.07	80.01	53.45	80.57	25.40	65.60	82.25
	U2PL [45]	**60.32**	64.55	**70.54**	72.57	**85.10**	79.43	53.88	80.17	26.24	65.87	82.43
	Ours	59.62	64.35	70.03	71.06	83.71	**81.69**	**54.94**	**81.38**	**27.93**	**66.08**	**82.60**
Deeplabv3+ [53]	Suponly	59.63	**65.75**	70.57	71.67	85.05	79.49	53.54	80.94	22.27	65.43	82.31
	MT [31]	59.79	65.47	70.68	71.96	84.54	80.09	54.54	80.61	27.15	66.09	82.55
	ClassMix [25]	59.88	65.41	71.28	72.37	85.14	80.75	54.77	80.43	27.23	66.36	82.75
	AEL [26]	59.23	64.65	72.99	72.32	**86.24**	**81.06**	53.77	80.92	26.74	66.44	82.67
	U2PL [45]	59.66	64.47	72.23	**72.86**	84.69	80.88	**54.94**	80.95	27.08	66.42	82.74
	Ours	**60.16**	65.70	**73.02**	72.48	85.82	80.41	53.39	**80.97**	**29.54**	**66.83**	**82.78**

* IS: Impervious surfaces. RB: Residential building. BL: Bare land.

TABLE X
COMPARISON WITH FULLY SUPERVISED METHODS ON THE AIR-SEG DATASET

Method	Network	IoU score per category(%)*									mIoU(%)	OA(%)
		IS	RB	Factory	Road	Water	Farmland	Grass	Tree	BL		
FSL	FCN [37]	59.58	63.95	69.78	71.45	82.61	78.78	53.10	80.50	24.02	64.86	81.88
	PSPNet [38]	60.09	64.50	68.89	71.81	84.82	80.66	53.10	80.87	25.98	65.64	82.35
	Deeplabv3 [39]	59.38	64.42	70.39	71.65	83.66	79.76	53.05	80.61	25.10	65.34	82.15
	Deeplabv3+ [53]	59.63	65.75	70.57	71.67	85.05	79.49	53.54	80.94	22.27	65.43	82.31
	DANet [59]	**60.40**	**65.76**	71.18	72.06	83.90	80.04	53.01	80.83	24.42	65.73	82.44
	UperNet [60]	59.73	65.42	70.50	71.58	84.50	80.31	53.61	81.02	24.89	65.73	82.47
	APCNet [61]	60.05	65.26	70.56	71.81	84.59	80.73	53.53	80.99	26.32	65.98	82.56
SSL	**Ours(Deeplabv3)**	59.62	64.35	70.03	71.06	83.71	**81.69**	**54.94**	**81.38**	27.93	66.08	82.60
	Ours(Deeplabv3+)	60.16	65.70	**73.02**	**72.48**	**85.82**	80.41	53.39	80.97	**29.54**	**66.83**	**82.78**

* IS: Impervious surfaces. RB: Residential building. BL: Bare land.

We validate the improvement for different models and comparison with fully supervised models.

1) Effectiveness of Our Method: The evaluation results are reported in Table IX. We conduct experiments on Deeplabv3 and Deeplabv3+ segmentation models. It can be seen that our method boosts 0.74% and 1.40% mIoU on Deeplabv3 and Deeplabv3+, respectively, compared to training with supervision only. Besides, our method surpasses the advanced semisupervised methods, especially on the bare land, which accounts for only 1.28% of the labeled dataset [9].

Fig. 13 shows the segmentation results for the AIR-SEG test set. Our approach causes the model to reduce the confusion between similar categories in the prediction results, such as farmland and grass. Moreover, we can obtain better segmentation results for road under large scenes using our method.

2) Comparison With Other Fully Supervised Models: We compare with excellent segmentation models that conduct fully supervised learning (FSL) based on the training set (29 images), including FCN [37], PSPNet [38], Deeplabv3 [39], Deeplabv3+ [53], DANet [59], UperNet [60], and APCNet [61] with ResNet50 as the backbone. As shown in Table X, our SSL approach achieves impressive precision compared to models of FSL. This proves that our method is able to effectively utilize the massive unlabeled images to extract knowledge from them to enhance the segmentation model.

V. CONCLUSION

In this article, we propose an effective semisupervised semantic segmentation framework based on label propagation and contrastive regularization for RSIs. The RCP module augments the unlabeled image with annotated class regions to achieve explicit label propagation. The PFC module constraints the pixel features of the unlabeled image using prototypes of labeled images to attain feature alignment through implicit label propagation. The RCL module performs contrastive regularization to learn invariant features under perturbations and compact class features on unlabeled images, which utilizes the structured graph of soft pseudo-labels to guide the relationship of pixel features. The effectiveness and versatility of our method have been validated on the ISPRS Vaihingen, Potsdam, and AIR-SEG datasets. Our framework substantially boosts the performance of the segmentation model under condition with a few annotations. Also, it outperforms the state-of-the-art semisupervised segmentation methods and can efficiently exploit many unlabeled images, surpassing the results of fully supervised segmentation models. With regard

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to future work, we would study on lightweight models to decrease the training and test time of our semisupervised segmentation model.

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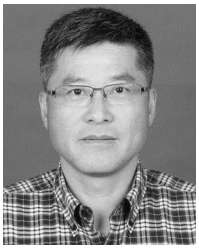
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